EXPERIMENT 1: STUDY OF NETWORK COMMANDS IN LINUX AND WINDOWS

AIM

To study network commands used in Linux and Windows and understand how they help inspect network configuration, connectivity, DNS, and routing.

1. HOSTNAME

Command: `hostname`

Sample output:

```
sabu-Mi-NoteBook-Ultra
```

Understanding: Displays the name assigned to the computer. This helps identify the host on a network.

2. IP ADDRESS CONFIGURATION (LINUX)

Command: `ip address show`

Sample output:

```
2: wlp2s0: <BROADCAST,MULTICAST,UP,LOWER_UP>
inet 192.168.1.105/24
link/ether 3c:xx:xx:xx:xx:xx
```

Understanding: Shows network interfaces, assigned IP addresses, and MAC addresses. The IP address identifies the device at the network layer; the MAC address identifies its network interface on the local link.

3. ARP CACHE

Command: `arp -a`

Sample output:

```
? (192.168.1.1) at aa:bb:cc:dd:ee:ff [ether] on wlp2s0
```

Understanding: Displays known mappings between local-network IPv4 addresses and MAC addresses. The entry may represent the router or another device recently contacted.

4. PING

Command: `ping -c 4 192.168.1.1`

Sample output:

```
64 bytes from 192.168.1.1: icmp_seq=1 ttl=64 time=1.2 ms
--- ping statistics ---
4 packets transmitted, 4 received, 0% packet loss
```

Understanding: Tests whether the destination responds to ICMP Echo Requests. Replies and 0% packet loss indicate successful reachability during this test; they do not guarantee that every service is available.

5. DNS LOOKUP

Command: `nslookup www.google.com`

Sample output:

```
Server: 192.168.1.1
Name: www.google.com
Address: 142.250.XXXX.XX
```

Understanding: Queries the Domain Name System (DNS) to resolve a domain name into IP address information. The exact server and address can vary.

6. ROUTING TABLE

Command: `ip route`

Sample output:

```
default via 192.168.1.1 dev wlp2s0
192.168.1.0/24 dev wlp2s0 proto kernel
```

Understanding: Displays routes used to forward IP packets. The default route sends traffic for destinations without a more specific route through the gateway.

7. NETWORK CONNECTIONS

Command: `netstat -r` (Windows or Linux, if available)

Sample output:

```
Network Destination Netmask Gateway
0.0.0.0 0.0.0.0 192.168.1.1
192.168.1.0 255.255.255.0 On-link
```

Understanding: The `-r` option displays the routing table. It helps identify the gateway and routes configured on the system.

PROCEDURE

1. Open Terminal in Linux or Command Prompt in Windows. 2. Enter each command and observe the displayed information. 3. Interpret the output to identify the host, IP/MAC details, connectivity, DNS resolution, and routing information. 4. Record the actual output obtained on the lab system.

RESULT

Thus, seven network commands were studied, and their sample outputs were interpreted to understand host identification, IP and MAC addressing, connectivity testing, DNS lookup, and routing.